



East West University
Department of Computer Science and Engineering

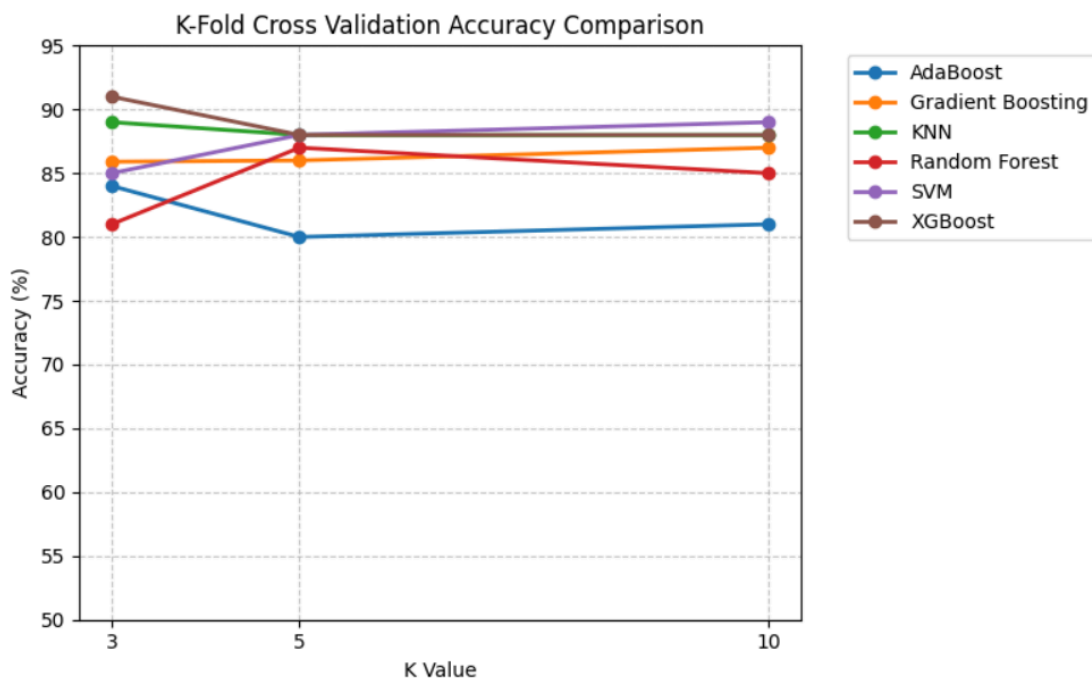
Course: CSE 475 (Machine Learning)
Section - 03

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Lab Report:04

K-fold cross-validation

Model	k=3	k=5	k=10
AdaBoost	84%	80%	81%
Gradient Boosting	85.9%	86%	87%
KNN	89%	88%	88%
Random Forest	81%	87%	85%
SVM	85%	88%	89%
XGBoost	91%	88%	88%



The K-Fold Cross Validation results demonstrate that all models achieved relatively high accuracy across different values of k, indicating good generalization performance.

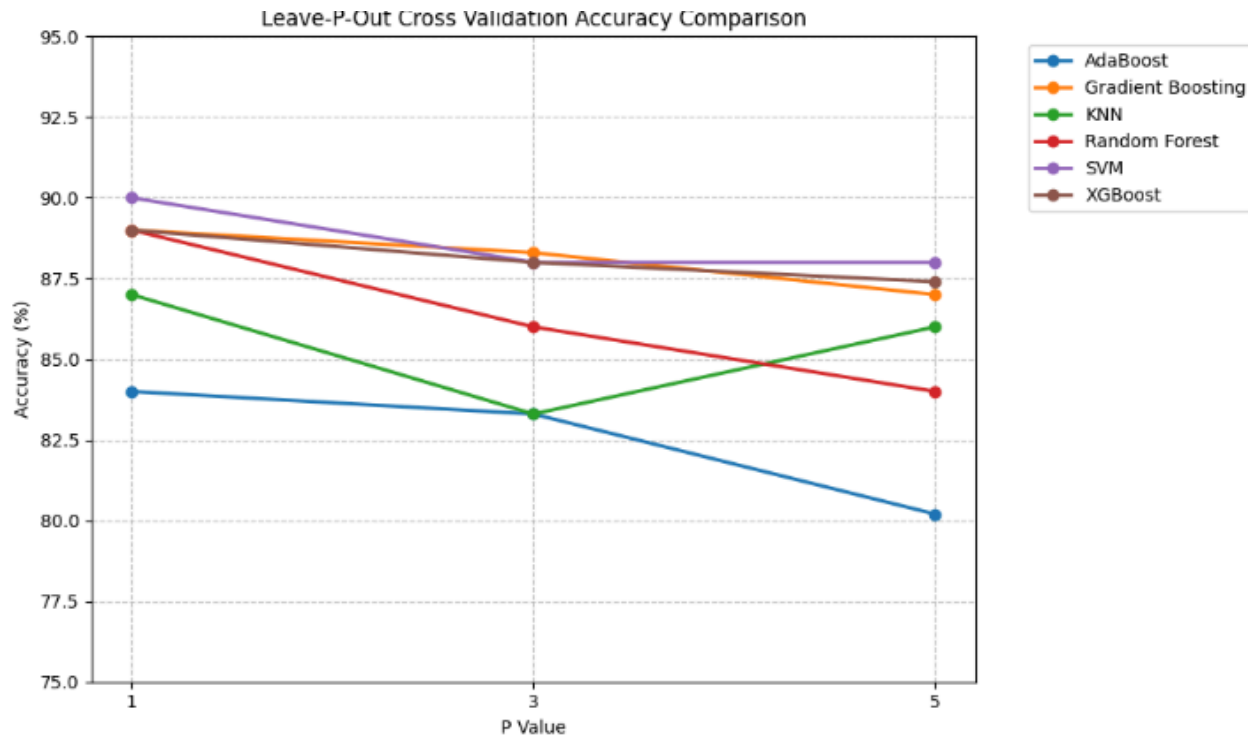
- **XGBoost** achieved the highest accuracy of 91% with k=3, making it the best-performing model overall. However, its accuracy slightly decreased to 88% for k=5 and k=10.

- **SVM** showed the most consistent improvement, increasing from 85% at k=3 to 89% at k=10, suggesting strong stability as the number of folds increased.
- **KNN** maintained consistently high performance, achieving 89%, 88%, and 88% for k=3, k=5, and k=10 respectively.
- **Gradient Boosting** demonstrated stable performance with accuracies ranging from 85.9% to 87%, indicating reliable predictive capability.
- **Random Forest** exhibited greater variation across folds, improving from 81% at k=3 to 87% at k=5 before slightly decreasing to 85% at k=10.
- **AdaBoost** produced the lowest performance among the evaluated models, with accuracies between 80% and 84%.

Overall, XGBoost achieved the highest peak accuracy, while SVM and KNN provided the most stable performance across different K-Fold configurations. Based on the results, XGBoost, SVM, and KNN are the strongest candidates for this classification

Leave-P-Out Cross-Validation

Model	p=1	p=3	p=5
AdaBoost	84%	83.3%	80.2%
Gradient Boosting	89%	88.3%	87%
KNN	87%	83.3%	86%
Random Forest	89%	86%	84%
SVM	90%	88%	88%
XGBoost	89%	88%	87.4%



Analysis

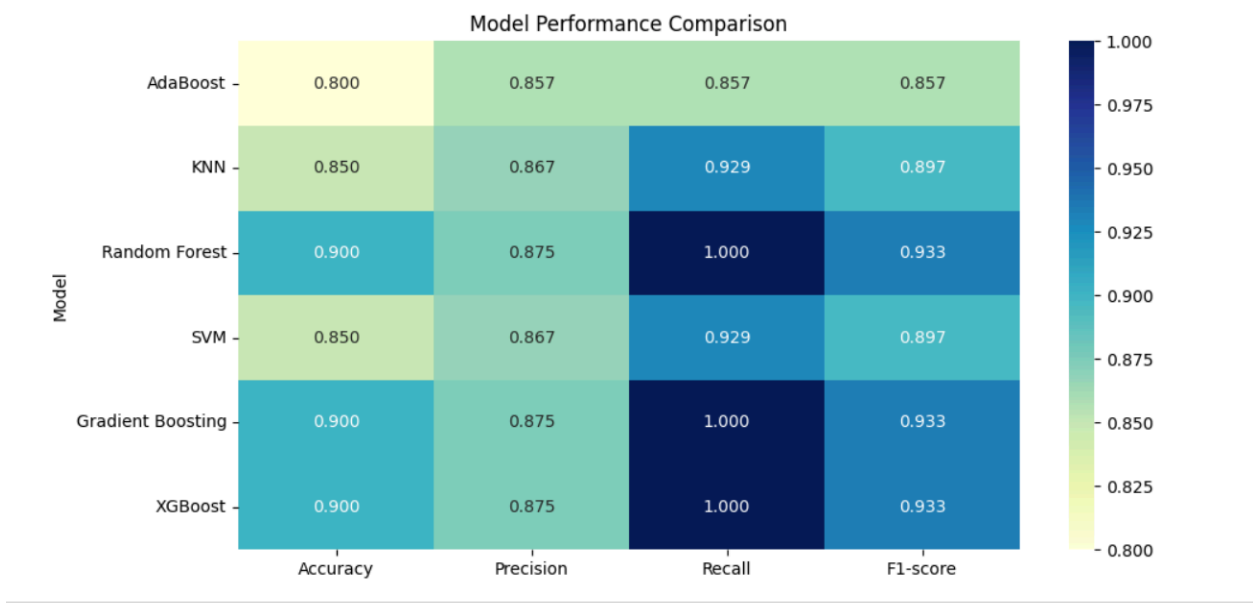
- **SVM** achieved the highest accuracy (**90%**) at $p=1$ and maintained strong stability at $p=3$ and $p=5$.
- **Gradient Boosting** and **XGBoost** performed consistently well, remaining close to 88–89% across all values of p .
- **Random Forest** showed a gradual decrease as p increased.
- **KNN** experienced a drop at $p=3$ but recovered at $p=5$.
- **AdaBoost** had the lowest performance and was the most affected by increasing p values.

Overall, **SVM** was the best-performing and most stable model under Leave-P-Out validation.

Key Findings

1. **XGBoost** achieved the highest overall accuracy (**91%**) in K-Fold validation.
2. **SVM** achieved the highest accuracy (**90%**) in Leave-P-Out validation.
3. KNN performed better in K-Fold than in Leave-P-Out.
4. Gradient Boosting and Random Forest benefited slightly from Leave-P-Out validation.
5. The performance differences between the two methods are small, suggesting that the models are stable and robust.

Overall Model Performance:



The heatmap visualization compares the performance of six machine learning algorithms using Accuracy, Precision, Recall, and F1-score. Random Forest, Gradient Boosting, and XGBoost achieved the best overall performance with 90% accuracy, 100% recall, and an F1-score of 93.3%. KNN and SVM demonstrated balanced performance with 85% accuracy and 89.7% F1-score. AdaBoost obtained the lowest accuracy of 80%, indicating comparatively weaker predictive capability. Overall, ensemble-based methods, particularly Random Forest, Gradient Boosting, and XGBoost, outperformed the other models and proved to be the most effective classifiers for the dataset.

Code:  Lab 4: Performance Metrics.ipynb